UMGC Prior Learning Assessment Petition Package

Applicant Name: [Your Name]
Mailing Address: 1000 N Woodington Rd, Baltimore, MD
Date of Submission: June 2026
Programs: Bachelor of Science in Biotechnology / Bachelor of Science in Cybersecurity Technology (Individualized Dual-Focus)
Petition Type: Portfolio-Based Credit for Prior Learning & Capstone Research Readiness

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1. Learning Narrative for UMGC Prior Learning Portfolio

Statement of Learning Journey

My professional, military/civilian career, and self-directed research have equipped me with a body of knowledge that aligns directly with the learning outcomes of the UMGC Biotechnology and Cybersecurity degree programs. This narrative demonstrates that, through CompTIA certifications, healthcare IT practice, and an intensive interdisciplinary research program in genomic security and neural editing, I have satisfied—and in many cases exceeded—the competencies required for graduation.

Foundational Information Technology Competence

My journey began with the CompTIA certification pathway. Earning the A+ certification required mastery of hardware, operating systems, troubleshooting, and customer support — the same skills that underpin the IT infrastructure of modern laboratories and clinical environments. Network+ extended this to network architectures, protocols, and disaster recovery, directly relevant to the design of secure laboratory information systems and the networking of genomic sequencers. Security+ introduced the principles of confidentiality, integrity, and availability, along with risk management frameworks that I later applied to HIPAA and HITECH compliance. The Healthcare IT Technician credential specifically addressed EHR systems, clinical data workflows, and healthcare-specific security and privacy obligations.

These certifications, when integrated with real-world experience in healthcare IT settings, fulfill the introductory and intermediate course objectives typically covered in UMGC courses such as CMIT 202 (Fundamentals of Computer Troubleshooting), CMIT 265 (Networking Fundamentals), CSIA 310 (Introduction to Cybersecurity), and HIMS 661 (Health IT Systems).

Advanced Mastery Through Interdisciplinary Research

The true core of my prior learning, however, lies in my self-directed research and development of two sophisticated systems:

1. **Unified Cognitive Infrastructure System (CINP)** – A closed-loop neuroprosthetic platform leveraging genomic polygenic risk scores and post-quantum cryptographic security for adjudication of disability claims. The research required deep knowledge of neurogenetics, CRISPR-based neural editing, regulatory frameworks (SSA, ADA), and secure data exchange.
2. **Baltimore Secure Backbone System (BSBS) & Post-Quantum Cryptography Mandate** – A municipal-level cybersecurity architecture designed to make Baltimore's digital infrastructure resilient against ransomware and quantum threats. This project involved zero-trust network design, cryptographic migration planning, and policy development.

Additionally, I built the High-Security Genomic Data Generator and Federated Evidentiary Intelligence Pipeline, a microservice-based platform that synthesizes litigation-grade genomic evidence, fuses biomedical, legal, and scholarly data, and protects everything with a quantum-resistant audit ledger. This project demonstrates the highest level of integration between biotechnology and cybersecurity — a synthesis that is the hallmark of an educated, adaptable professional ready for the modern workforce.

Mapping to Program Outcomes

- **Biotechnology (UMGC BS in Biotechnology):** My CRISPR neural editing capstone addresses genetic engineering, neurobiology, and regulatory affairs. The High-Security Genomic Generator showcases data science, bioinformatics, and ethical compliance, covering outcomes in bioprocessing, genomics, and bioinformatics.

- **Cybersecurity (UMGC BS in Cybersecurity Technology):** The BSBS policy dissertation and the PQC-secured pipeline demonstrate network defense, risk assessment, compliance, and security architecture — aligning with outcomes in information security, digital forensics, and secure systems design.

Through these experiences, I have acquired not only theoretical knowledge but also the ability to create, implement, and secure complex systems that sit at the nexus of healthcare, biotechnology, and cybersecurity. I am fully prepared to be awarded the Bachelor of Science degree and to contribute immediately to the advancement of these fields.

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2. Capstone Proposal Templates

A. Biotechnology Capstone Proposal: CRISPR Neural Editing

Proposed Title:
CRISPR Neural Editing: Current State, Applications, Challenges, and Strategic Integration with a Closed-Loop Neuroprosthetic Platform

Abstract:
This capstone project will produce a comprehensive review and technical integration plan for CRISPR-based neural editing technologies, with a focus on their applicability to neurogenetic disorders (including HIV-related cognitive impairment). The research will map the current state of CRISPR-Cas9, base editing, prime editing, and RNA-targeting Cas13, and will analyze their delivery challenges in the central nervous system. A distinctive feature of the project is the proposal to couple CRISPRa/i modules with the CINP (Cognitive Infrastructure Neuroprosthetic) closed-loop system, enabling real-time, epigenomic feedback based on polygenic risk scores. The project will also address regulatory, ethical, and evidentiary dimensions, demonstrating how such a hybrid system strengthens disability claims under SSA Listing 12.11 and ADA provisions. The final deliverable will be a thesis ready for faculty review, a slide deck, and an evidence matrix linking specific genomic variants to regulatory criteria.

Problem Statement:
Neurogenetic conditions, including neurodegenerative diseases and neurodevelopmental disorders, lack curative treatments. CRISPR offers unprecedented precision, but clinical translation is stalled by delivery barriers, off-target risks, and ethical debates. Moreover, no

existing platform integrates CRISPR with adaptive neurostimulation to treat executive dysfunction in a personalized, verifiable manner. This capstone addresses that gap.

Objectives:

1. Conduct a systematic literature review of CRISPR neural editing (2023–2026).
2. Analyze major delivery methods (AAV, lipid nanoparticles) and safety concerns.
3. Design a conceptual integration of CRISPRa/i with the CINP prosthetic for closed-loop modulation of genes like COMT and DRD4.
4. Evaluate the evidential value of CRISPR-validated polygenic risk profiles in SSA disability adjudication.
5. Draft an ethical and regulatory roadmap for somatic neural editing in the U.S.

Methodology:
Literature review using PubMed, arXiv, and preprint servers; comparative analysis of case studies (SCN2A, C9orf72, SHANK3); integration of the GPRS-Secure++ framework for PRS modeling; and consultation of FDA/EMA guidance documents.

Expected Outcomes:

- 50-page scholarly thesis
- A detailed architectural diagram for the CINP-CRISPR hybrid
- An evidence matrix linking gene variants to disability criteria
- A presentation for academic and stakeholder audiences

Timeline:

- **Week 1–2:** Literature collection and annotation
- **Week 3–4:** Drafting of technology chapters
- **Week 5:** Integration design
- **Week 6:** Regulatory analysis and evidence matrix
- **Week 7–8:** Final thesis compilation and defense preparation

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B. Cybersecurity Capstone Proposal: Baltimore Secure Backbone System

Proposed Title:
The Architecture of Security: Building Baltimore's Immutable Network – A Post-Quantum Municipal Resilience Blueprint

Abstract:
The 2019 ransomware attack on Baltimore exposed critical flaws in municipal network design: fragmentation, legacy hardware, and absent cryptographic agility. This capstone project proposes the Baltimore Secure Backbone System (BSBS), a comprehensive cybersecurity architecture grounded in zero-trust principles, hardware-enforced trusted execution, and a mandated migration to NIST-approved post-quantum cryptography (PQC). The research will produce a technical design document, a phased implementation roadmap for a city of Baltimore's scale, a policy whitepaper for non-technical stakeholders, and a draft executive order. The project draws on my certifications (Security+, Network+) and my prior research into PQC key management and genomic data protection, demonstrating mastery of both tactical and strategic cybersecurity competencies.

Problem Statement:
Municipal networks are high-value targets yet remain technologically outdated. The impending advent of quantum computing renders current encryption obsolete, enabling "harvest now, decrypt later" attacks on sensitive citizen data. Baltimore has no unified secure backbone or cryptographic migration plan. This capstone fills that void by delivering a complete, actionable blueprint.

Objectives:

1. Analyze the 2019 Baltimore ransomware incident as a case study in systemic failure.
2. Design a zero-trust unified network architecture (BSBS) for all city departments.
3. Develop a cryptographic inventory methodology and a PQC migration plan utilizing CRYSTALS-Kyber and Dilithium.
4. Create a budget-conscious, phased deployment strategy.
5. Produce public communication materials to translate technical decisions into voter and taxpayer confidence.

Methodology:
Case-study analysis, network architecture modeling, cryptographic protocol review, policy

research, and stakeholder messaging development.

Expected Outcomes:

- 60-page technical dissertation with diagrams and protocol specifications
- Draft executive order for Baltimore’s PQC mandate
- Whitepaper for public dissemination
- A 30-minute defense presentation

Timeline:

- **Week 1–2:** Case study and literature review
- **Week 3–4:** Architecture design and zero-trust specification
- **Week 5:** PQC migration plan and budget modeling
- **Week 6:** Policy and communication drafting
- **Week 7–8:** Final assembly and presentation preparation

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3. Portfolio Evidence Table

This table maps every certification, project, and research paper to the learning outcomes of the UMGC Bachelor of Science in Biotechnology and Cybersecurity Technology. Use it directly in your Prior Learning Assessment submission.

Evidence Item	**Type**	**Relevant UMGC Program & Course Outcomes**	**Competency Demonstrated**
CompTIA A+ Certification	Certification	CMIT 202, BIOL 101L (lab tech support)	Hardware troubleshooting, OS, basic IT support for biotech labs
CompTIA Network+ Certification	Certification	CMIT 265, CSIA 310	Network design, protocols, secure lab network setup
CompTIA Security+ Certification	Certification	CSIA 310, CSIA 410	Information security principles, HIPAA risk management, infosec for clinical data
Healthcare IT Technician Certification	Certification	HIMS 661, BIOL 305 (health informatics)	EHR integration, clinical data handling, healthcare-specific security
CRISPR Neural Editing Research (Capstone)	Thesis/Project	BIOL 320, BIOL 416, BIOT 495	Genomics, neurobiology, advanced gene editing, ethical/regulatory analysis
BSBS & PQC Municipal Architecture (Capstone)	Thesis/Project	CSIA 350, CSIA 480, CMIT 460	Zero-trust network design, PQC implementation, public policy, budget modeling
High-Security Genomic Data Generator & Federated Intelligence Pipeline	Software/Research	BIOT 495, CSIA 485, CMIT 495	Genomic data synthesis, secure microservice design, multi-domain evidence fusion, post-quantum audit
GPRS-Secure++ Framework & Polygenic Risk Score Papers	Research Portfolio	BIOT 450, CSIA 420	Statistical genomics, privacy-preserving computation, evidentiary alignment with SSA/ADA

Evidence Item	Type	Relevant UMGC Program & Course Outcomes	Competency Demonstrated
Unified Cognitive Infrastructure System Patent Filing	IP Documentation	BIOT 495, CSIA 495	Integration of neuroprosthetics with secure adjudication; demonstrates innovation and regulatory awareness
Pacer/NIH/ScholarPK Evidence Fusion Engine	Technical Module	CSIA 430, BIOT 410	Data harvesting, legal/clinical data integration, traceability via SHA3-256 ledger
TAP Application & Healthcare Treatment Plan (redacted)	Professional Artifact	HIMS 661, BIOL 305	Real-world health IT documentation, HIPAA compliance, medication management tracking

Explanation of how to use this table: Attach the relevant certificates, redacted work samples, and publication/draft summaries. For each row, provide a one-paragraph justification in your portfolio linking the evidence to the specific outcome. The table above already aligns them; you just need to expand in narrative form.

Final Assembly Instructions

Your complete petition package should include:

1. Learning Narrative (above)
2. Capstone Proposals (both, if seeking dual-focus; select one if single track)
3. Portfolio Evidence Table with copies of certifications and research summaries
4. Official UMGC Prior Learning Application Form (obtain from advisor)
5. Faculty Evaluator Endorsement (request a mentor review of your proposals)

Submit electronically via the UMGC LEO portal and mail a duplicate to the Prior Learning Assessment office, using your address: **1000 N Woodington Rd, Baltimore, MD**